PROPRIETARY & CONFIDENTIAL — LWANDILE ENGINEERING PROJECTS & INCLUSIVE ENTREPRENEURIAL ACADEMY

SCAPUSH PRECISION STANDARD

SECTION 02 — EXECUTIVE SUMMARY

Sovereign Bio-Solar-Thermal Campus | Master Design Specification

Doc Ref: 02_executive_summary.html | Status: APPROVED BASELINE | Governance: Scapush Precision | JET Window 4 Aligned

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1. Executive Lock-In Register

LISTEN SECTION

This section codifies the mandatory high-level project parameters locked under Master Design Specification `MDS-001` for official Just Energy Transition (JET) evaluation:

Ref Code	Locked Core Parameter	Design Specification & Scope Boundary
ES-001	Facility Identity	Food-Energy-Water Nexus Demonstration Campus
ES-002	Site Boundary	20,000 m^2 (2 Hectares) Leasehold Site with 565m Secured Perimeter
ES-003	Poultry Cluster	1 Barn (12m × 32m) | 4 Biosecure Pods | 4,800 to 5,720 Broilers Capacity
ES-004	Swine Cluster	1 Barn (12m × 32m) | 8 Pens in 4 Pods | 160 to 500 Weaners (Feedstock Role)
ES-005	Energy Philosophy	100% Off-Grid Sovereignty (Independent of Utility Grid)
ES-006	Energy Architecture	Integrated Solar PV + Battery Storage (BESS) + Biogas CHP Engine
ES-007	Prohibited Energy	Zero Diesel Generators (Strictly Excluded from Design Basis)
ES-008	Accessibility Baseline	Accessibility-First Infrastructure (Universal Access, EV/Mobility Hub)
ES-009	Skills Partner	Inclusive Entrepreneurship Academy (Dr. Palesa Charlotte Felix-Faure)
ES-010	Impact Target	JET-Aligned Circular Economy & Disability Beneficiation Platform

2. The Opportunity & Project Overview

LISTEN SECTION

The Socioeconomic Opportunity

South Africa's Just Energy Transition presents an opportunity to demonstrate how food security, renewable energy, accessibility and entrepreneurship can be integrated into a single regenerative platform.

Many communities across the Mpumalanga coal belt face simultaneous challenges relating to:

- Food insecurity & volatile supply chains
- Energy insecurity & frequent grid collapse
- Unemployment & coal value chain contraction
- Disability exclusion & physical barriers
- Skills shortages in clean technology
- Waste management challenges
- Limited participation in the green economy

Project Overview

The **Sovereign Bio-Solar-Thermal Campus** is a proposed off-grid Food-Energy-Water Nexus Demonstration Facility integrating livestock production, renewable energy, biogas generation, thermal recovery, digital monitoring, skills development and inclusive entrepreneurship.

The campus is designed as a living demonstration of how circular economy principles can support food production, energy generation and community development within a single infrastructure platform.

3. Resource Circulation & Differentiation

LISTEN SECTION

CIRCULAR ECONOMY CORE PHILOSOPHY

Unlike conventional agricultural projects, the campus is designed around resource circulation rather than resource consumption. Each subsystem directly supports and feeds into another subsystem.

Waste → Biogas
LIVESTOCK MANURE BECOMES DIGESTER FEEDSTOCK

Biogas → Power
BIOGAS CHP GENERATES FIRM ELECTRICITY

Heat → Biology
CHP WASTE HEAT MAINTAINS DIGESTER/BARN MICROCLIMATES

Digestate → Crop
DIGESTATE PROVIDES HIGH-GRADE ORGANIC BIO-FERTILIZER

Operations → Skills
DAILY CAMPUS OPERATIONS PROVIDE PRACTICAL TRAINING

Skills → Ventures
ACADEMY TRAINING FOSTERS COMMUNITY ENTREPRENEURSHIP

4. Technical Infrastructure & Energy Platform

LISTEN SECTION

Biosecure Production Clusters

Poultry Production Cluster:

Energy Sovereignty Platform

- One 12m × 32m Poultry Barn
- Four Biosecure Pods (4,800 to 5,720 Broiler Capacity)
- Microclimate Nursery & Climate Moderation
- SCADA Digital Telemetry Integration

Swine Nursery Cluster:

- One 12m × 32m Swine Barn
- Four Biosecure Pods / 8 Pens (160 to 500 Weaner Capacity)
- Primary Organic Feedstock Generation Role for Anaerobic Digester
- SCADA Digital Telemetry Integration

The facility is designed around permanent off-grid operation supported by three integrated systems:

- **Solar PV Array:** Primary daytime power generation and battery charging.
- **Battery Energy Storage (BESS):** Lithium Iron Phosphate (LFP) storage for night and peak load leveling.
- **Biogas Combined Heat & Power (CHP):** Firm thermal and electrical energy support driven by anaerobic digestion.

MANDATORY PROHIBITION:
Diesel generators form no part of the design basis under any operational condition.

5. Accessibility Leadership & Entrepreneurship Academy

LISTEN SECTION

Accessibility Infrastructure Mandate

Accessibility is treated as a core infrastructure requirement, ensuring complete participation for persons with disabilities:

- Universal Step-Free Access Routes across 100% of campus.
- Accessible Learning Spaces & Administration Facilities.
- Dedicated Mobility Wheelchair & Scooter Charging Bays.
- Off-grid DC EV Accessibility Charging Hub.
- Inclusive Skills Development Programmes tailored for all abilities.

Inclusive Entrepreneurship Academy

Partnered with Dr. Palesa Charlotte Felix-Faure, the Academy provides accredited green economy training and venture development:

- Renewable Energy Systems & Solar Maintenance
- Biogas Operation & Anaerobic Digestion
- Controlled Agriculture & Digital Farming
- Livestock Systems & Biosecurity Management
- SCADA & ArIoT Telemetry Operations
- Green Enterprise & Cooperative Governance

6. JET Alignment & Executive Declaration

LISTEN SECTION

Just Energy Transition Alignment

- Renewable Energy Adoption
- Green Skills Development
- Economic Inclusion & Decent Jobs
- Climate Resilience & Adaptation
- Accessibility Inclusion Leadership
- Circular Economy Demonstration

- Community Participation & Ownership
- Food Security Enhancement

EXECUTIVE DECLARATION

"The Sovereign Bio-Solar-Thermal Campus is not merely an agricultural project. It is a demonstration of how biological systems, renewable energy systems, digital technologies and community participation can be integrated into a single regenerative infrastructure platform capable of creating lasting social, environmental and economic value."

7. Approval & Authorization

LISTEN SECTION

PROJECT STEWARD & TECHNICAL LEAD



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Sipho Wiseman Khuzwayo

Inclusive Legacy Architect / Principal Lead

Lwandile Engineering Projects (Pty) Ltd

Status: Digitally Sealed (2026-10-23 22:18:17 UTC) [Scapush Precision]

GOVERNANCE & SKILLS PARTNER

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Dr. Palesa Charlotte Felix-Faure

Founder & Academic Lead

Inclusive Entrepreneurial Academy

Status: Awaiting Signature File

Consortium Leadership

Sipho Wiseman Khuzwayo — Lead System Architect (Lwandile Engineering)
Dr. Palesa Charlotte Felix-Faure — Academic & Governance Lead (Inclusive Entrepreneurial Academy)

Execution Entities & Standard

Lwandile Engineering Projects (Pty) Ltd
Inclusive Entrepreneurial Academy
Scapush Precision Execution Standard

Operational Footprint

Sovereign Bio-Solar-Thermal Campus, Mpumalanga, South Africa

Primary Contact & Communication

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